

UI/UX DESIGN DOCUMENT

PhishShield

A Machine Learning-Based Web Application for Phishing URL Detection

1. Introduction

The UI/UX Design Document describes how users will interact with the PhishShield web application and how the interface communicates the phishing-detection result. The design follows the documented workflow: a user enters a website URL, URL-based features are processed by a trained machine-learning model, the URL is classified as Legitimate or Phishing, and a security recommendation is presented.

2. Objectives of UI/UX Design

Provide a simple interface for entering and checking a URL.

Make the prediction result immediately visible and understandable.

Provide a clear security recommendation after analysis.

Reduce unnecessary steps between URL entry and result.

Maintain consistent visual hierarchy across screens.

Support responsive desktop and mobile browser use.

3. Design Principles

USER EXPERIENCE AND INFORMATION ARCHITECTURE

4. Target Users

5. User Journey

6. Information Architecture

7. Main User Flow

Figure 1. Main user flow of the PhishShield interface.

WIREFRAMES AND SCREEN DESIGN

8. Home / URL Checker Screen

The home screen is the primary interaction point. It should immediately communicate the purpose of the application and provide one prominent URL input field and one primary action.

Figure 2. Low-fidelity wireframe for the main URL checker.

9. Result Screen

The result area should make the prediction visually prominent, followed by a short explanation and security recommendation. The design should not rely on color alone; the status text must explicitly state the result.

Figure 3. Low-fidelity wireframe for the prediction result.

10. UI Element Specification

VISUAL DESIGN SYSTEM

11. Typography

The document follows the academic visual language of the supplied Database Design Document: Times New Roman, strong serif headings, restrained blue/navy accents, tables, and a clean page structure. The web application can use a web-safe equivalent while preserving the same hierarchy.

12. Color and Visual Language

13. Layout Guidelines

Use a centered content area with sufficient white space.

Keep the URL checker visually dominant.

Use consistent margins, spacing and button sizes.

Place prediction and recommendation above secondary information.

Use responsive stacking on small screens.

Avoid unnecessary technical details in the main user flow.

14. Interaction States

RESPONSIVE DESIGN AND ACCESSIBILITY

15. Responsive Design

PhishShield is intended for browser use. The interface should adapt to desktop, tablet and mobile screens without changing the core workflow.

16. Accessibility Requirements

Provide a visible label for the URL input.

Allow keyboard focus and form submission.

Use readable text and sufficient contrast.

Do not communicate phishing status using color alone.

Provide text alternatives for icons.

Show validation/error messages near the relevant control.

Keep touch controls large enough on mobile.

17. Usability Considerations

18. Security-Aware UX

Because PhishShield is a cybersecurity application, the interface must avoid creating a false sense of certainty. Results should clearly communicate that the system provides a machine-learning-based assessment. The UI should not expose unnecessary technical or sensitive information.

19. Feedback Messages

USABILITY TESTING AND DESIGN VALIDATION

20. Usability Testing Plan

The interface should be evaluated with representative users before final implementation. Testing focuses on whether users understand the purpose, submit a URL correctly, interpret the prediction, and understand the recommendation.

21. Evaluation Criteria

Task completion rate.

Time required to submit and understand a result.

Number of input or navigation errors.

User understanding of Legitimate vs Phishing.

User satisfaction with clarity and simplicity.

Accessibility and responsive-layout issues.

22. Design Validation

The final UI will be validated after frontend-backend integration. The team will verify that the result displayed by the interface matches the prediction returned by the backend and that loading, success, invalid-input and error states behave consistently.

23. Team Responsibilities

FUTURE ENHANCEMENTS AND CONCLUSION

24. Future UI/UX Enhancements

Browser extension interface for direct URL checking.

Scan history interface if enabled in a future database version.

Visualization of model confidence and contributing URL indicators.

Security analytics dashboard.

Dark mode and additional accessibility options.

Mobile-first interface or dedicated mobile application.

Educational explanations for common phishing indicators.

25. UI/UX Design Summary

The proposed PhishShield interface is centered on one understandable journey: enter a URL, request analysis, wait for processing, and interpret the prediction with a security recommendation. The design minimizes navigation, emphasizes clear feedback, supports responsive layouts, and incorporates accessibility and security-aware communication.

26. Conclusion

The UI/UX design provides a structured interface for the PhishShield web application while maintaining the project's core phishing-detection workflow. The proposed screens, user flow, visual hierarchy, interaction states, responsive behavior, accessibility requirements and usability testing plan establish a practical foundation for frontend implementation.

27. Alignment with Existing Project Documentation

This UI/UX document is aligned with the supplied PhishShield Database Design Document. That document specifies the workflow in which users enter a website URL, URL-based features are processed by a trained machine-learning model, the URL is classified as Legitimate or Phishing, and a security recommendation is provided. It also identifies scan time and scan information as application data. These workflow elements are reflected in the proposed UI and interaction design.

Document Item	Details
Application Type	Web Application
Primary User Action	Enter a website URL and request analysis
Primary Output	Legitimate / Phishing prediction with security recommendation
Frontend	HTML, CSS, JavaScript
Backend	Python Flask
Machine Learning	Scikit-learn
Database	MySQL
Design Focus	Simple, clear, trustworthy and responsive

Principle	Application
Simplicity	Focus the interface on URL submission and result viewing.
Clarity	Use familiar labels such as URL, Check URL, Legitimate and Phishing.
Consistency	Use consistent buttons, typography, spacing and layouts.
Feedback	Show processing state and clear success/error results.

Principle	Application
Trust	Avoid exaggerated claims; explain that the result is a model prediction.
Accessibility	Use readable text, keyboard-friendly controls and adequate contrast.

User Group	Needs
General Internet User	Quickly assess whether a URL appears suspicious.
Student / Learner	Understand phishing detection and the result.
Security-Conscious User	Check suspicious URLs and view guidance.

Stage	User Action	System Response
1. Discover	Open PhishShield	Home screen explains purpose.
2. Input	Paste/type URL	Input accepts URL.
3. Submit	Select Check URL	Validation and analysis begin.
4. Analyze	Wait	Processing feedback is shown.
5. Understand	View prediction	Result and recommendation appear.
6. Decide	Follow guidance	User decides whether to proceed or avoid.

Level	Screen / Element	Purpose
Primary	Home / URL Checker	Main entry point for URL analysis
Primary	Result Area	Shows classification and recommendation
Supporting	About / Information	Explains phishing detection and project purpose
Supporting	Help / Guidance	Provides basic instructions

User	Interface	Backend / ML
Enter URL	URL input field	—
Click Check URL	Validation + loading	Request received
—	Processing state	Feature extraction
—	Processing state	ML prediction
—	Result screen	Prediction returned
Read recommendation	Result + guidance	Optional scan record stored

PHISHSHIELD — URL CHECKER
Enter website URL
https://example.com
Enter a URL to check for phishing indicators.
[CHECK URL]
Result area appears after analysis

URL ANALYSIS RESULT
Submitted URL: example.com
Prediction: LEGITIMATE / PHISHING
Confidence: optional

URL ANALYSIS RESULT
Security Recommendation: follow safe browsing guidance
[CHECK ANOTHER URL]

Element	Design / Behavior
URL Input	Large field with label, placeholder and validation.
Check URL	Primary action; prevent repeated submission while processing.
Loading State	Short processing message or spinner.
Prediction	Prominent Legitimate or Phishing label.
Recommendation	Short actionable security guidance.
Error Message	Clear explanation without technical jargon.
Navigation	Minimal navigation keeps detection central.

Text Type	Recommended Treatment
Page Title	Large, bold, high-contrast heading
Section Heading	Bold heading with clear spacing
Body Text	Readable regular text with comfortable line height
Button Text	Short, bold action label
Result Status	Large, bold label plus explicit status text

Element	Suggested Treatment	Purpose
Primary Brand	Deep navy / blue	Professional and trustworthy appearance
Primary Action	Blue button	Clear main action
Legitimate Result	Positive indicator + text	Communicate classification
Phishing Result	Warning indicator + text	Communicate risk clearly
Background	Light neutral	Reduce distraction
Cards / Panels	White surface + subtle border	Separate content areas

State	UI Behavior
Initial	URL field and Check URL button visible.
Invalid Input	Show validation message near the input.
Loading	Show processing feedback and prevent duplicate submission.
Legitimate	Show classification and recommendation.
Phishing	Show classification prominently with warning guidance.
Error	Explain that analysis failed and provide retry guidance.

Screen	Layout Approach
Desktop	Centered content with URL checker and result panel.
Tablet	Flexible content width and spacing.
Mobile	Single-column layout; full-width input, button and stacked result.

Goal	Design Decision
Fast task completion	One main input and one main action.

Goal	Design Decision
Low learning curve	Familiar wording and minimal navigation.
Result comprehension	Classification and recommendation shown together.
Error recovery	Allow correction and retry.
User confidence	Explain that the result is an ML-based assessment.

Situation	Example Message
Empty URL	Please enter a URL to analyze.
Invalid URL	Please enter a valid website URL.
Processing	Analyzing the URL. Please wait.
Phishing	Potential phishing URL detected. Avoid entering sensitive information on this site.
Legitimate	No phishing indicators were detected by the model. Continue normal safe-browsing practices.
Error	The URL could not be analyzed right now. Please try again.

Test	Task	Expected Result
UT-01	Open application	Purpose is clear.
UT-02	Enter valid URL	URL is accepted.
UT-03	Submit URL	Loading state is visible.
UT-04	Read legitimate result	User identifies result and recommendation.
UT-05	Read phishing result	User recognizes warning and guidance.
UT-06	Enter invalid input	Clear validation appears.
UT-07	Use on mobile	Core task works without horizontal scrolling.

Role	UI/UX Responsibility
Frontend Developer	Implement screens, responsive layout, components, validation and interaction states.
Backend Developer	Provide API responses, validation behavior and integration support.
Machine Learning Engineer	Define prediction outputs and confidence/explanation information shown in the UI.
Cybersecurity & Testing	Review security-aware messaging, test usability, validate error states and security.